ATHARVA SHINDE

Buffalo, NY | 716-247-0475 | shindeatharva0324@gmail.com | linkedin.com/in/atharva-shinde-51b646284 | github.com/atharva0324 | atharva0324.github.io

SUMMARY

Master's in Data Science candidate (Dec 2026) with hands-on experience building LLM- and agent-based systems, from tool-use and retrieval-augmented generation to schema-validated, production-grade outputs. Comfortable working across the AI agent stack LangChain/LangGraph orchestration, structured LLM outputs with Pydantic, RAG pipelines, and REST API integration deployed with Docker, FastAPI, MLflow, and AWS. Seeking an entry-level AI/ML or Agent Engineer role building diagnostic and optimization agents in a production, closed-loop environment.

TECHNICAL SKILLS

Generative AI & Agents: LangChain, LangGraph, Agentic AI, Agent Orchestration, Tool-Use, MCP-style Tool Ecosystems, Prompt Engineering, Retrieval-Augmented Generation (RAG), Structured/Schema-Validated Outputs (Pydantic), Vector Embeddings & Vector Databases, HuggingFace Transformers, Sentence Transformers, LLM Fine-Tuning, LLM APIs (OpenAI)

Languages & Engineering: Python, R, SQL, typed/data-modeled code (Pydantic, TypedDict), REST API integration, Git/GitHub

ML & DL: Scikit-learn, XGBoost, TensorFlow, Keras, PyTorch, Neural Networks, CNNs, RNNs, LSTMs, Transformers, BERT, NER, Random Forest, Clustering, PCA, Hyperparameter Tuning (Optuna), Feature Engineering, Cross-Validation, A/B Testing

MLOps, Deployment & Evaluation: Docker, FastAPI, MLflow (experiment tracking & offline evaluation), Streamlit, AWS EC2, CI/CD (GitHub Actions), Model Serialization, Inference Pipelines

Data & BI: MySQL, PostgreSQL, Power BI, Pandas, NumPy, Matplotlib, Seaborn, Jupyter

PROJECTS

JobFit AI — Semantic Resume & Job-Description Matching Agent
Stack: NLP, BERT, Sentence Transformers, FastAPI, MLflow, HuggingFace
github.com/atharva0324/JobFit-AI

- Built a hybrid scoring pipeline (80% semantic similarity via sentence-transformers, 20% keyword match) to resolve synonym mismatches, improving candidate-to-JD matching accuracy by 60%.
- Fine-tuned BERT (bert-base-uncased) on 14.5K sentences for NER-based skill extraction, reaching 0.50 entity-level F1, and surfaced actionable, auditable skill-gap feedback for each match.
- Deployed the end-to-end system on HuggingFace Spaces with FastAPI and Streamlit, with MLflow tracking for reproducible offline evaluation of model versions.

YouTube Video RAG Chatbot — Tool-Augmented Conversational Q&A
Stack: LangChain, LCEL, Chroma, OpenAI Embeddings, FastAPI, Streamlit, Python
github.com/atharva0324/Youtube_chatbot

- Built a retrieval-augmented chatbot over YouTube transcripts, answering queries in under 5 seconds and cutting manual video search time by 1–3 hours per video.
- Implemented a conditional tool-use pipeline (RunnableBranch) that falls back to a Wikipedia tool only when transcript context is insufficient, using PydanticOutputParser to validate sufficiency checks and structure outputs.
- Added multi-turn conversational memory (ChatPromptTemplate, MessagesPlaceholder), supporting 6+ sequential questions per session, exposed via a FastAPI backend with a Streamlit interface.

PCOS Risk Prediction System — Monitored Production ML API
Stack: Python, XGBoost, FastAPI, Docker, AWS EC2, MLflow, Streamlit, Optuna
github.com/atharva0324/pcos-api

- Shipped an XGBoost classification API (0.80 recall on the positive class) via FastAPI, Dockerized and deployed on AWS EC2 with a Streamlit front end.
- Built observability into the training loop: benchmarked Random Forest, SVM, and XGBoost across 5+ MLflow-logged experiments, tracking 10+ metrics per run (precision, recall, thresholds) to catch regressions before model promotion.
- Automated testing and deployment with a 100% GitHub Actions CI/CD pipeline, triggered on every push to main.

WORK EXPERIENCE

SIES Graduate School of Technology — Data Science Intern *Jun 2024 – Jul 2024*

- Built an end-to-end data pipeline for an Indian flight-pricing dataset, integrating 3+ sources with automated cleaning scripts, cutting data prep time by 30%.
- Performed exploratory data analysis to uncover key pricing patterns, then engineered predictive features that improved Random Forest and XGBoost cross-validation performance by 8%.
- Validated models using k-fold cross-validation and statistical metrics (precision, recall, F1), ensuring results were reliable before translating them into pricing insights.
- Delivered a full Airlines Price Analysis project, translating model results into actionable, business-readable pricing insights for stakeholders.

EDUCATION

University at Buffalo, SUNY — Master's in Data Science *Aug 2025 – Dec 2026*

SIES Graduate School of Technology — B.E. in Artificial Intelligence & Machine Learning *Jul 2021 – May 2025*